

Mathematics – Grade 6

Fractions

Introduction

A fraction represents a part of a whole. It consists of two numbers:

- **Numerator** – the number above the line.
- **Denominator** – the number below the line.

Example:

$$\frac{3}{4}$$

Here:

- Numerator = 3
 - Denominator = 4
-

Types of Fractions

Proper Fractions

A proper fraction has a numerator smaller than the denominator.

Examples:

- $\frac{2}{5}$
 - $\frac{3}{8}$
 - $\frac{7}{10}$
-

Improper Fractions

An improper fraction has a numerator greater than or equal to the denominator.

Examples:

- $\frac{7}{5}$
- $\frac{9}{4}$

- $8/8$
-

Mixed Numbers

A mixed number contains a whole number and a fraction.

Examples:

- $2 \frac{1}{3}$
 - $4 \frac{5}{6}$
-

Equivalent Fractions

Equivalent fractions represent the same value.

Examples:

- $1/2 = 2/4 = 4/8$

Multiply or divide the numerator and denominator by the same number to obtain equivalent fractions.

Comparing Fractions

When fractions have the same denominator, compare the numerators.

Example:

$$3/8 > 1/8$$

When denominators are different, convert them into equivalent fractions.

Example:

$$1/2 = 3/6$$

$$2/3 = 4/6$$

Therefore,

$$2/3 > 1/2$$

Adding Fractions

Same Denominator

Add the numerators and keep the denominator the same.

Example:

$$1/5 + 2/5 = 3/5$$

Different Denominators

Find the Least Common Denominator (LCD).

Example:

$$\begin{aligned} 1/2 + 1/4 \\ = 2/4 + 1/4 \\ = 3/4 \end{aligned}$$

Subtracting Fractions

Example:

$$5/6 - 2/6 = 3/6 = 1/2$$

Multiplying Fractions

Multiply numerators together.

Multiply denominators together.

Example:

$$\begin{aligned} 2/3 \times 4/5 \\ = 8/15 \end{aligned}$$

Dividing Fractions

Keep the first fraction.

Change division to multiplication.

Flip the second fraction.

Example:

$$2/3 \div 4/5$$

$$= 2/3 \times 5/4$$

$$= 10/12$$

$$= 5/6$$

Real-Life Applications

Fractions are used in:

- Cooking recipes
- Measuring ingredients
- Sharing food
- Reading maps
- Construction
- Shopping discounts